Week 4: Data Visualization & Capstone Project

End-to-End Retail Sales Analysis | Explanation Report

Dataset note: The course handout references a Google Sheets dataset, but the uploaded handout does not contain its URL or data. This report therefore uses a synthetic practice dataset included in the ZIP. Replace it with the official dataset if required.

1. Objective

Demonstrate the complete analyst workflow: load data, clean records, explore sales, aggregate metrics, visualize patterns, and communicate recommendations to a stakeholder.

2. Dataset and fields

The sample contains order ID, order date, region, product, units, unit price, and revenue. It intentionally includes one missing region and a repeated order ID to demonstrate cleaning.

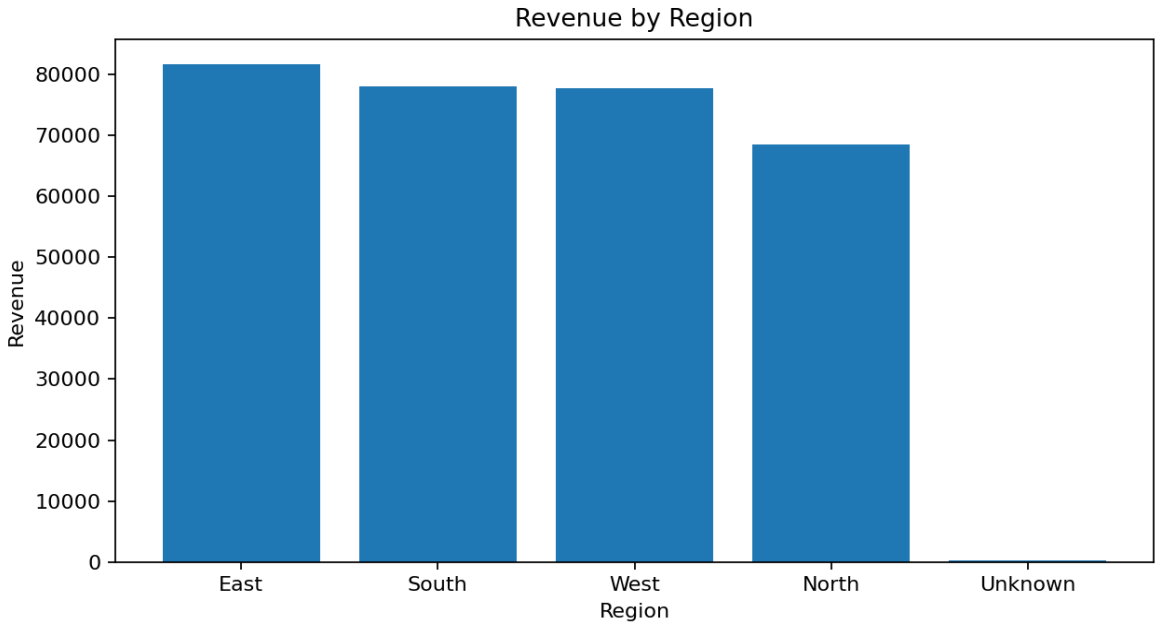
3. Methodology

Cleaning: remove duplicate order IDs, replace missing region with “Unknown”, remove rows missing required values, and recompute revenue as units × unit price.
Analysis: calculate descriptive statistics, revenue by region, revenue by product, and monthly revenue.
Visualization: create bar charts for region and product performance and a line chart for monthly trend.

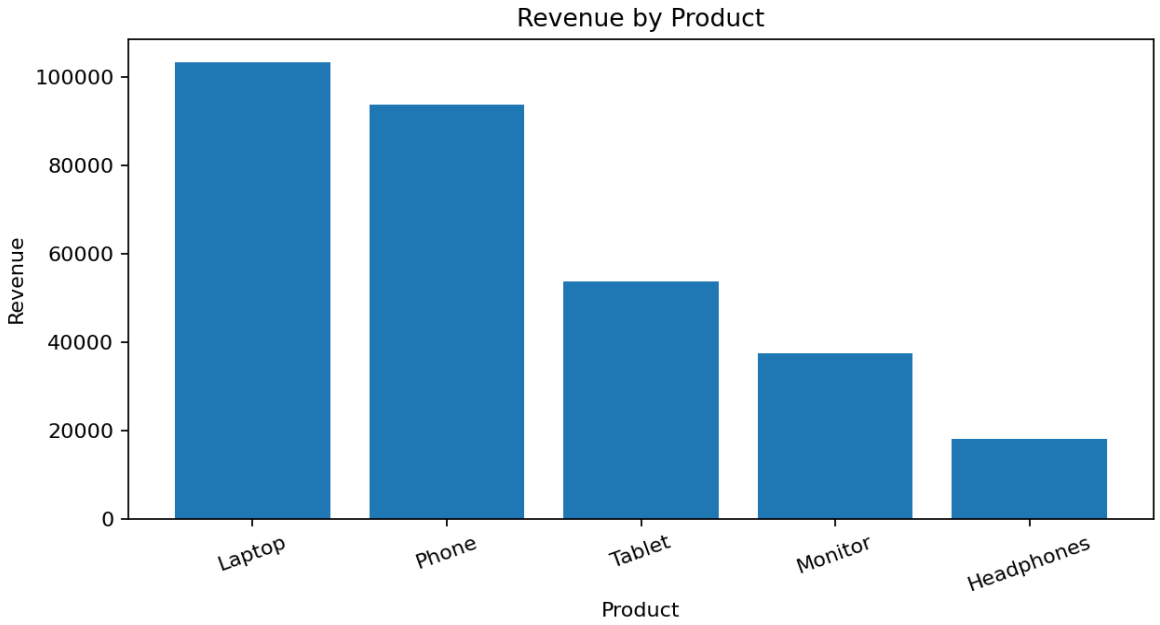
4. Results

After removing the repeated order ID, the analysis includes 240 orders and total revenue of 306,170 (sample currency units). The highest-revenue region is East (81,705); the highest-revenue product is Laptop (103,350).

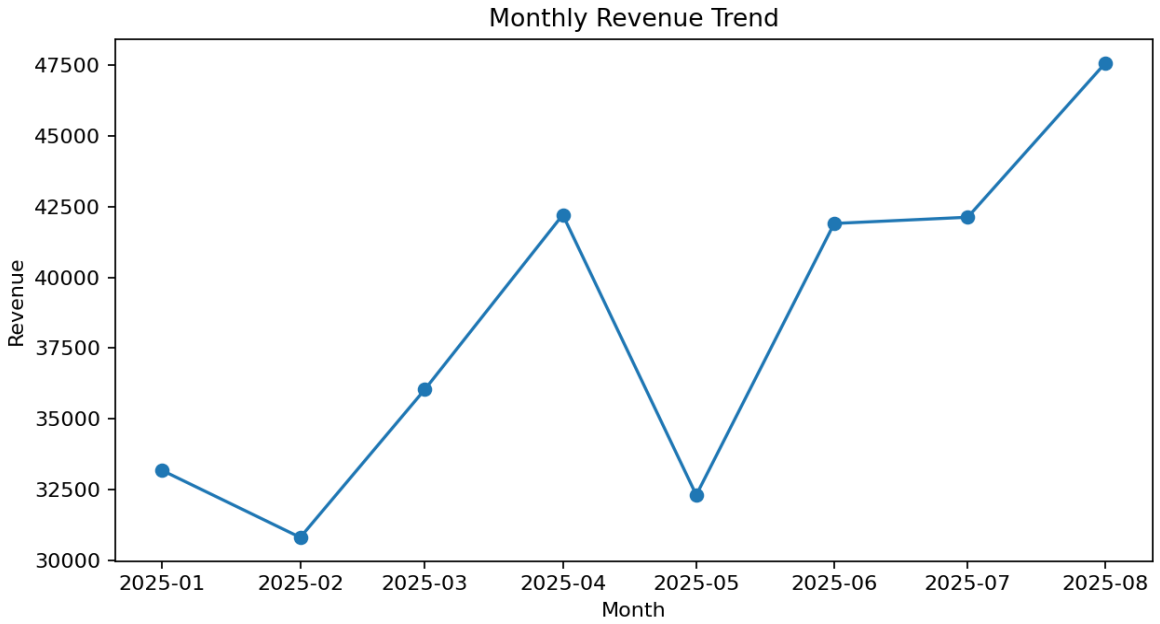
Revenue by Region



Revenue by Product



Monthly Revenue Trend



5. Stakeholder summary and recommendations

Use the region and product rankings to prioritize inventory and marketing. Review monthly revenue for periods of growth or decline, then investigate causes using campaign, pricing, and stock data. Treat the findings as illustrative because the dataset is synthetic. In Power BI or Tableau, build KPI cards for revenue, units, and order count, add region/product/date filters, and place the three charts on one dashboard.

6. Reproducibility

The ZIP contains the CSV, Python script, requirements file, README, and generated output tables/charts. Run `pip install -r requirements.txt` followed by `python analysis.py`.

7. Limitations

The sample is generated for learning and cannot support real business decisions. No official Google Sheets dataset URL was present in the uploaded course handout. Replace the sample data and rerun the script for course submission.